Aditya Ghosh Curriculum Vitae

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Contact details

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Address: Computing and Data Science (CoDa)
389 Jane Stanford Way, Stanford University
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Education

Ph.D. in Statistics, **Stanford University** GPA 4.13/4.00 2022–2027

- Advised by Stefan Wager (Stanford GSB) & Dominik Rothenhäusler (Stanford Statistics)

Masters of Statistics (M.Stat), **Indian Statistical Institute**, Kolkata 2020–2022

- **Dissertation advisor:** Prof. Bodhisattva Sen (Columbia University)

Bachelor of Statistics (B.Stat), **Indian Statistical Institute**, Kolkata 2017–2020

Research

A central goal of my research is to design robust methods for causal inference in complex settings that arise in modern applications, such as dynamic systems that evolve over time, units interacting with shared resources, or heterogeneity across different locations and time periods. I aim to build practical tools that accommodate these richer structures, while being as transparent and rigorous as the standard toolkit.

5. **Ghosh, A.** & Wager, S. (2025+). Non-parametric Causal Inference in Dynamic Thresholding Designs. *Major revision at the Journal of the Royal Statistical Society Series B: Statistical Methodology; extended abstract accepted at ACM EC 2026* arXiv
4. **Ghosh, A.** & Rothenhäusler, D. (2025+). Which Covariates to Adjust for? Specification-robust Causal Inference in Observational Studies. *Major revision at the Annals of Statistics.* arXiv R package
3. **Ghosh, A.**, Imbens, G. & Wager, S. (2025+). PLRD: Partially Linear Regression Discontinuity Inference. *Major revision at the Proceedings of the National Academy of Sciences (contributed).* arXiv R package
2. **Ghosh, A.**, Deb, N., Karmakar, B., & Sen, B. (2026). Robustness and Efficiency of Rosenbaum's Rank-based Estimator in Randomized Trials: A Design-based Perspective. *Accepted at Biometrika.* arXiv Journal link
1. **Ghosh, A.** (2019). An asymptotic formula for the Chebyshev theta function. *Notes on Number Theory and Discrete Mathematics*, 25(4), 1-7. Journal link

Ongoing

6. **Ghosh, A.**, Rothenhäusler, D. & Xu, Y. Random Distribution Shifts in Panel Data
7. **Ghosh, A.**, Wager, S. & Ashlagi, I. Long-term Policy Evaluation in Deceased-donor Kidney Allocation

Academic services

- I help organize the **Online Causal Inference Seminar**, a weekly seminar on causal inference with participants from all over the world.
- Served as a reviewer for: *AoS*, *JASA*, *J. Econom.*, *JRSSB*, *NeurIPS*

Teaching

As instructor, **Stanford University** ExploreCourses

- Stats 302: Qualifying Exam Workshop (Theoretical Statistics). Summer 2025, 2026
- Stats 302: Qualifying Exam Workshop (Probability). Summer 2024

As teaching assistant, **Stanford University** ExploreCourses

- Stats 361: Causal Inference. Winter 2025
- Stats 310B/Math 230B: Theory of Probability II. Winter 2024
- Stats 310A/Math 230A: Theory of Probability I. Autumn 2023
- Stats 216: Introduction to Statistical Learning. Winter 2023
- Stats 209: Introduction to Causal Inference. Fall 2025
- Stats 203: Regression Models and Analysis of Variance. Fall 2026
- Stats 202: Data Mining and Analysis. Summer 2023, Autumn 2022
- Stats 200: Introduction to Theoretical Statistics. Winter 2026, Autumn 2024
- Stats 141: Introduction to Statistics for Biology. Spring 2026
- Stats 60: Introduction to Statistical Methods: Precalculus. Spring 2025

Other experiences

- Trained numerous high school students for mathematics olympiads and other math competitions; materials are available at ghoshadi.wordpress.com .

Accolades

- **Departmental Teaching Assistant Award**, Stanford University 2026
- **American Statistical Association San Francisco Bay Area Chapter**, student travel award for JSM 2026 2026
- **IMS 2025 International Conference on Statistics and Data Science (ICSDS)**, student travel award 2025
- **IISA conference 2025**, best poster award 2025
- **Industrial Affiliates Annual Conference at Stanford**, travel award 2024
- Selected for **Machine Learning in Economics Summer Institute** at University of Chicago Booth school of business 2024
- **ISIAA–J. K. Ghosh Memorial Gold Medal** (for outstanding performance in Masters of Statistics) 2023
- **ISIAA–Mrs. M. R. Iyer Memorial Gold Medal** (for best overall performance in Bachelor of Statistics) 2021
- **Nikhilesh Bhattacharyya Memorial Gold Medal** (for best performance in major (Statistics) in Bachelor of Statistics) 2021
- **Madhava Mathematics Competition**, selected among the top few participants in the country for a prestigious summer camp 2019, 2018
- **Indian National Mathematical Olympiad**, earned a prestigious certificate of merit from NBHM, Govt. of India (awarded to the top 75 participants in the country) 2016

Class Projects

- **SMARTer Multi-task Fine-tuning of BERT**. Joint with Disha Ghandwani and Rahul Kanekar.
- **Inference for Data Adaptively Sampled via REINFORCE**. Joint with Ivy Zhang.
- **Predicting the Number of Micro-businesses Across the U.S.** Joint with Joshua Kazdan and Yash Nair.

Skills

Python, **R**, Markdown, Git, C (basic), HTML (basic).

R packages: plrd specrobust

Invited Talks

- **2026 California Econometrics Conference** in Honor of Guido Imbens 2026
  Title: Partially Linear Regression Discontinuity Inference.
- Women and Allies in Statistics (WAIS) Research Talk Workshop 2026
  Title: Non-parametric Causal Inference in Dynamic Thresholding Designs
- **Joint Statistical Meeting**, Boston, Massachusetts 2026
  Title: Which Covariates to Adjust for? Specification-robust Causal Inference in Observational Studies.
- **ACM Conference on Economics and Computation (EC'26)**, Rome, Italy 2026
  Title: Nonparametric Causal Inference in Dynamic Thresholding Designs
- **Metrics-lunch** (an weekly seminar on Econometrics), Stanford University 2026
  Title: Non-parametric Causal Inference in Dynamic Thresholding Designs
- **Data-driven Seminar**, Stanford University 2026
  Title: Robustness and Efficiency of Rosenbaum's Rank-based Estimator in Randomized Trials: A Design-based Perspective
- **University of Washington** Causal Inference & Missing Data Reading Group 2026
  Title: Assumption-robust Causal Inference
- **IMS International Conference on Statistics and Data Science**, Seville, Spain 2025
  Title: Assumption-robust Causal Inference
- **Stanford Causal Science Center (SC^2) Conference**, Stanford University 2025
  Title: Causal Inference in Dynamic Thresholding Designs
- **Joint Statistical Meeting**, Nashville, Tennessee 2025
  Title: Causal Inference in Dynamic Thresholding Designs
- **Industrial Affiliates Annual Conference**, Stanford University 2024
  Title: Practical Bias-aware Inference in Regression Discontinuity Designs
- **Stanford Causal Science Center (SC^2) Conference**, Stanford University 2024
  Title: Asymptotic bias-aware inference in regression discontinuity designs under higher-order smoothness
- **Metrics-lunch** (an weekly seminar on Econometrics), Stanford University 2024
  Title: Practical Bias-Aware Inference for Regression Discontinuity: An Asymptotic View
- **Computational and Methodological Statistics**, Berlin, Germany 2023
  Title: Efficiency and robustness of Rosenbaum's regression (un)-adjusted rank-based estimator in randomized experiments
- **PCM Memorial Lecture**, Indian Statistical Institute, Kolkata 2022
  Title: The synthetic control method in causal inference
- **D. Basu Memorial Lecture**, Indian Statistical Institute, Kolkata 2021
  Title: Large low-rank matrix completion
- **Online Reading Group on Functional Data Analysis** 2021
  Title: Two-sample testing of the equality of mean functions
- **Students' Learning Seminar**, Indian Statistical Institute, Kolkata 2021
  Title: Matching estimators in causal inference

References

Prof. Stefan Wager
Stanford Graduate School of Business
Ph.D. co-advisor
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Prof. Guido Imbens
Stanford Graduate School of Business
Research collaborator
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Prof. Dominik Rothenhäusler
Department of Statistics, Stanford University
Ph.D. co-advisor
rdominik@stanford.edu

Prof. Bodhisattva Sen
Department of Statistics, Columbia University
Research collaborator
bodhi@stat.columbia.edu